IILM UNIVERSITY, GREATER NOIDA
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
LANGUAGE THEORY AND AUTOMATA
B.TECH (CSE) V SEM

Assignment-1

Submission Date: 3 September 2026
Note: Assignment must be submitted handwritten only to the concern faculty.

Q1: Define alphabet, string, empty string, length of a string, and language with suitable examples. Let Σ= {0, 1}

Construct the following languages:

- L1 ={w|w contains exactly two 1s}
- L2= {w|w starts and ends with the same symbol}
  Write at least **five strings** belonging to each language.

Q2: Design a **DFA over Σ= {0, 1}** that accepts all strings containing **at least one occurrence of the substring** `101`.

- Draw the state diagram.
- Write the transition table.
- Test DFA for the strings `101`, `1101`, `1001`, and `10101`.

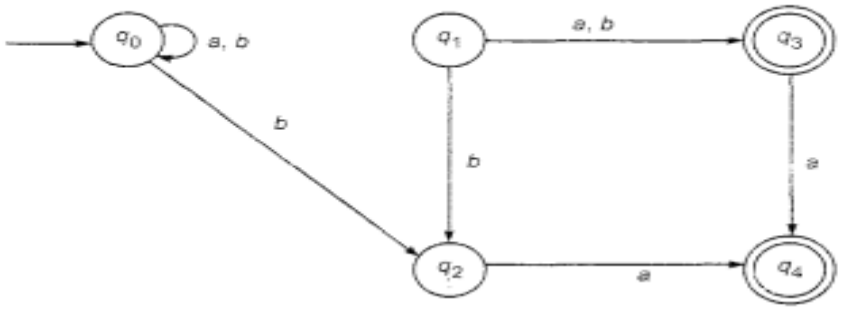
Q3: Find a DFA equivalent to

M = ({ q_0,q_1,q_2}, (0,1), δ, q_0, q_2) where δ is given by the table

State	Input	
	0	**1**
→q_0	q_0,q_1	q_2
q_1	q_0	q_1
q_2 (Final State)	--	q_0,q_1

Q4:

Construct a DFA equivalent to the NDFA M whose transition diagram is given by Fig.



Q5: Construct a DFA accepting all strings over {a, b} ending in ab.

Q6: Construct a minimum state automaton equivalent to following transition diagram.

State	**Input**	
	0	1
→q_0	q_1	q_2
q_1	q_1	q_3
q_2	q_3	q_4
q_3	q_1	q_5
q_4	q_4	q_2
q_5(final state)	q_5	q_5

Q7:

Consider the finite state machine whose transition function δ is given by Table in the form of a transition table. Here, $Q = \{q_0, q_1, q_2, q_3\}$, $\Sigma = \{0, 1\}$, $F = \{q_0\}$. Give the entire sequence of states for the input string 110001.

TABLE Transition Function Table

State	Input	
	0	1
→ q_0	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2

Q8:

Prove that if $\delta(q, x) = \delta(q, y)$, then $\delta(q, xz) = \delta(q, yz)$ for all strings z in Σ^*.

Q9:

Using the **Pumping Lemma**, prove that the language

$$L = \{0^n 1^n \mid n \geq 0\}$$

is **not regular**.

Q10:

Consider the DFA below:

- States: {q0, q1}
- Start State: q0
- Final State: q1

Transitions:

Current State	Input 0	Input 1
q0	q0	q1
q1	q1	q1

Using **Arden's Theorem**, derive the corresponding **Regular Expression** accepted by this DFA.